Properties of equality

1 Write the new equation produced for each of the following scenarios:

a Adding 1 to both sides of $x = 9$

b Subtracting 7 from both sides of $x = 14$

c Multiplying 5 to both sides of $x = 10$

d Dividing 11 to both sides of $x = 99$

e Adding 4 to both sides of $x = -6$

f Subtracting 20 from both sides of $x = 2$

g Multiplying 8 to both sides of $x = -3$

h Dividing 5 to both sides of $x = -25$

i Multiplying 11 to both sides of $5x = 6$

j Multiplying 12 to both sides of $3x = -8$

k Dividing 3 to both sides of $9x = 27$

l Dividing 9 to both sides of $45x = -54$

m Multiplying $\frac{7}{8}$ to both sides of $5m = 3$

n Multiplying $\frac{1}{3}$ to both sides of $-4x = 5$

2 Determine whether the following equations are equivalent:

a $-5y = 13$ and $\frac{-5y}{-5} = \frac{13}{5}$

b $\frac{u}{6} = 4$ and $6 \times \frac{u}{6} = 4$

3 Determine the property that justifies why the equations $9x = -27$ and $\frac{9x}{9} = \frac{-27}{9}$ are equivalent.

4 Fill in the blank so that each resulting statement is true.

a The addition property of equality states that if $a = b$, then $a + c = \square$.

b The multiplication property of equality states that if $a = b$ and $c \neq 0$, then $ac = \square$.

5 Consider the equation $3\,(a + (-6)) + 10\,(2a + 5) = 8$. Determine the property that justifies every step.

a $3a + (-18) + 20a + 50 = 8$ is obtained after the first step.

b $-18 + 50 + 3a + 20a = 8$ is obtained after the second step.

c $(-18 + 50) + (3a + 20a) = 8$ is obtained after the third step.

6 A student is in the process of solving an equation. The original equation and the first step are as follows:

$$\text{Original: } 9a + 7 = 2 - 5a + 6$$
$$\text{Step 1: } 9a + 7 = 2 + 6 - 5a$$

Determine the property used by the student for the first step.

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7 Consider the following equations:

$$\text{Equation 1: } 3x = -15$$
$$\text{Equation 2: } \frac{3x}{3} = \frac{-15}{3}$$

a Determine the property why Equation 1 and Equation 2 are equivalent.

b State the number that is multiplied to both sides of Equation 1 resulted to Equation 2.

8 Determine whether the following are valid deductions from the statement $a = 12$:

a $2a = 6$ **b** $a + 7 = 19$ **c** $\frac{a}{2} = 6$ **d** $a + 1 = 12$

9 Determine the property illustrated for each statement.

a If $a = d$, then $a + 3 = d + 3$.

b Given that $b = 16$, then $b - 5 = 11$.

10 Consider the equation $21 = x + 13$.

a Solve for x.

b Kathleen started to solve the equation as follows:

$$21 = x + 13$$
$$21 - 13 = x + 13 - 13$$

Determine the property that justifies this step of work.

c Kathleen finished solving the equation as shown:

$$21 - 13 = x + 13 - 13$$
$$8 = x + 0$$
$$8 = x$$

Determine the property that justifies Kathleen's final step of work.

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11 Consider the equation $7x = 49$.

a Solve for x.

b Ray started to solve the equation as follows:

$$\begin{aligned} 7x &= 49 \\ \frac{7x}{7} &= \frac{49}{7} \end{aligned}$$

Determine the property that justifies this step of work.

c Ray finished solving the equation as shown:

$$\begin{aligned} \frac{7x}{7} &= \frac{49}{7} \\ 1x &= 7 \\ x &= 7 \end{aligned}$$

Determine the property that justifies Ray's final step of work.

12 Consider the equation $t - 14 = 0$.

a Solve for t.

b Vanessa started to solve the equation as follows:

$$\begin{aligned} t - 14 &= 0 \\ t - 14 + 14 &= 0 + 14 \end{aligned}$$

Determine the property that justifies this step of work.

c Vanessa finished solving the equation as shown:

$$\begin{aligned} t - 14 + 14 &= 0 + 14 \\ t - 0 &= 0 + 14 \\ t &= 14 \end{aligned}$$

Determine the property that justifies Vanessa's final step of work.

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13 Consider the equation $\dfrac{8x+48}{9}=16$.

a Solve for x.

b Dylan started to solve the equation as follows:

$$\frac{8x+48}{9}=16$$
$$8x+48=144$$

Determine the property that justifies Dylan's first step of work.

c Dylan continued to solve the equation as shown:

$$8x+48=144$$
$$8x=96$$

Determine the property that justifies this step of work.

d Dylan finished solving the equation as follows:

$$8x=96$$
$$x=12$$

Determine the property that justifies Dylan's final step of work.

14 Consider the equation $\dfrac{2x-4}{3}=4$.

a Solve for x.

b Skye solved the equation as follows:

$$\text{Given equation: } \frac{2x-4}{3}=4$$
$$\text{Step 1: } 2x-4=12$$
$$\text{Step 2: } 2x=16$$
$$\text{Step 3: } 1x=8$$
$$\text{Solution: } x=8$$

Determine the property that justifies Skye's final step of work.

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15 Consider the equation $\frac{4(x+6)}{5} = 20$.

a Solve for x.

b Susana started solving the equation as follows:

$$\frac{4(x+6)}{5} = 20$$
$$4(x+6) = 100$$

Determine the property that justifies Susana's first step of work.

c Susana continued to solve the equation as shown:

$$4(x+6) = 4$$
$$x+6 = 1$$

Determine the property that justifies this step of work.

d Susana finished solving the equation as follows:

$$x+6 = 1$$
$$x = -5$$

Determine the property that justifies Susana's final step of work.

16 Consider the equation $\frac{2(x-6)}{9} = 6$.

a Solve for x.

b Ned solved the equation as follows:

$$\begin{aligned}
\text{Given equation:} \quad & \frac{2(x-6)}{9} = 6 \\
\text{Step 1:} \quad & \frac{2(x-6)}{9} \times 9 = 6 \times 9 \\
\text{Step 2:} \quad & 2(x-6) = 54 \\
\text{Step 3:} \quad & \frac{2(x-6)}{2} = \frac{54}{2} \\
\text{Step 4:} \quad & x-6 = 27 \\
\text{Step 5:} \quad & x-6+6 = 27+6 \\
\text{Step 6:} \quad & x-0 = 33 \\
\text{Solution:} \quad & x = 33
\end{aligned}$$

Determine the property that justifies Ned's final step of work.

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17 Consider the equation $4\left(\frac{x}{6}+7\right)=8$.

a Solve for x.

b Sally started to solve the equation as follows:

$$4\left(\frac{x}{6}+7\right)=8$$
$$\frac{x}{6}+7=2$$

Determine the property that justifies Sally's first step of work.

c Sally continued to solve the equation as shown:

$$\frac{x}{6}+7=2$$
$$\frac{x}{6}=-5$$

Determine the property that justifies this step of work.

d Sally finished solving the equation as follows:

$$\frac{x}{6}=-5$$
$$x=-30$$

Determine the property that justifies Sally's final step of work.

18 Consider the equation $6\left(\frac{x}{9}-4\right)=18$.

a Solve for x.

b Irene solved the equation as follows:

$$\text{Given equation: } 6\left(\frac{x}{9}-4\right)=18$$
$$\text{Step 1: } \frac{6\left(\frac{x}{9}-4\right)}{6}=\frac{18}{6}$$
$$\text{Step 2: } \frac{x}{9}-4=3$$
$$\text{Step 3: } \frac{x}{9}-4+4=3+4$$
$$\text{Step 4: } \frac{x}{9}=7$$
$$\text{Step 5: } \frac{x}{9}\times 9=7\times 9$$
$$\text{Step 6: } 1x=63$$
$$\text{Solution: } x=63$$

Determine the property that justifies Irene's final step of work.

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